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A Introduction

Goal of the work, maybe a simple example, some mention to preliminary information and a soft mention to what's the problem and the philosophy.

B Minkowski space and space-like surfaces

The environment we will work within is the Minkowki 3-space, defined as $\mathbb{R}^3$ endowed with the pseudometric

$$ds^2 = (dx^1)^2 + (dx^2)^2 - (dx^3)^2.$$

In order to distinguish the use of this metric from the standard euclidean metric of $\mathbb{R}^3$ we will use the notation $\mathbb{M}^{2,1}$. We will also denote the coordinates either as (x^1, x^2, x^3) or (x, y, t), choosing the notation according to convenience. We will also identify the slice $\{(x, y, 0)\}$ of $\mathbb{M}^{2,1}$ with $\mathbb{R}^2$ and with $\mathbb{C}$.

The space $\mathbb{M}^{2,1}$ is an example of a more general object called a pseudo-Euclidean vector space, a term that refers to a finite-dimensional vector space V endowed with a nondegenerate (possibly not positive-definite) symmetric bilinear form g. This kind of space serves as the "infinitesimal model" for pseudo-Riemannian manifolds, smooth manifolds endowed with a smooth nondegenerate $(0, 2)$-tensor. If the signature of this tensor is exactly $(n-1, 1)$, the manifold is called a Lorentzian manifold.
The following definition, borrowed from physics, will be useful later.

Definition. Given a Lorentzian manifold (M, g), we say that a vector $v \in T_pM$ is

- space-like if $\langle v, v\rangle > 0$
- light-like if $\langle v, v\rangle = 0$
- time-like if $\langle v, v\rangle < 0$

We will always denote with S a differentiable manifold of dimension 2 (a surface) immersed in $\mathbb{M}^{2,1}$, $\iota : S \hookrightarrow \mathbb{M}^{2,1}$. As usual, the immersion induces a

scalar product on S, the first fundamental form or pull-back metric, which we indicate as $\mathbf{I} = \iota^*(g_{\mathbb{M}^{2,1}})$ and write in coordinates as $\mathbf{I} = g_{ij}dx^i dx^j$ (for simplicity we always forget the immersion ι).

Space-like immersions.
The object we will focus on are not arbitrary immersed surfaces but the so-called space-like surfaces of $\mathbb{M}^{2,1}$. This term refers to the immersed surfaces in $\mathbb{M}^{2,1}$ whose induced metric is positive-definite hence Riemannian, or in other terms, they're surfaces S whose tangent bundle $\{(p, v)\}$ is made only of space-like vectors.

A nice property of this kind of surface is that they are locally the graph of a function of the form $x^3 = f(x^1, x^2)$.
We can show this with a simple argument:
First, we assumed that the pseudometric restricted to S,$\langle\cdot,\cdot\rangle|_S \geq 0$, is definite positive, which means that the tangent vectors of S at any point p belong to the set of space-like vectors:

$$T_p \subset \mathscr{C} := \big\{v \in \mathbb{M}^{2,1} : \langle v, v\rangle \geq 0\big\} = \{t^2 \leq x^2 + y^2\}.$$

This is just the outside-region of a double cone in $\mathbb{M}^{2,1}$, whose symmetry axis is $\big\{x^1 = 0, x^2 = 0\big\}$. A plane is entirely made of space-like vectors iff it's contained between the two branches of the cone.
Then, having assumed that S is an immersed surface, hence an immersed submanifold, it is locally the solution of an equation of the form $F(x, y, z) = 0$. It is well known that from this local expression we can express the normal vector to the surface at a point p: $N_p = \vec{\nabla}F(p)$.
Furthermore, we observe that N_p is orthogonal to the space-like plane T_pS and so it cannot be horizontal. The last two statements taken together give us $\partial_z F(p) = (N_p)^3 \neq 0$, and so we can apply the implicit function theorem and express the surface as the graph of $z = f(x, y)$ near the point p.
Where the local representation is valid, we can parametrize the surface as $(x, y, f(x, y))$, and derive the local frame $V_x = (1, 0, \partial_x f(x, y))$, $V_y = (1, 0, \partial_y f(x, y))$, for which the first fundamental form is represented by the matrix

$$g = \begin{pmatrix} \langle V_x, V_x\rangle & \langle V_y, V_x\rangle \\ \langle V_x, V_y\rangle & \langle V_y, V_y\rangle \end{pmatrix} = \begin{pmatrix} 1 - f_x^2 & -f_x f_y \\ -f_x f_y & 1 - f_y^2 \end{pmatrix}.$$

Since the parametrization is obviously regular, the condition on g being a metric translates in having a strictly positive determinant,

$$0 < \det g = 1 - (f_x^2 + f_y^2) = 1 - |\vec{\nabla}f|^2,$$

which means $|\vec{\nabla}f|^2 < 1$.

Entire surfaces.
Given a surface S immersed in $\mathbb{M}^{2,1}$, we can define a projection $\Pi : S \to \mathbb{R}^2$ by forgetting the time coordinate, $\Pi(x^1, x^2, x^3) := (x^1, x^2)$ for any $(x^1, x^2, x^3) \in S$. We call the surface S is entire when the map Π is surjective. If the surface is globally a graph of a function φ, this condition is exactly stating that the

function φ is an entire map from $\mathbb{R}^2$ to $\mathbb{R}$.

The situation for space-like surfaces is actually a lot nicer than what we presented up to now, as the representation as a graph is not only local but actually global as we prove in the proposition below.

Proposition 1. *Let Ω be a domain in $\mathbb{C}$ (that is, an open connected set) and let $S : \Omega \to \mathbb{M}^{2,1}$ be a space-like immersion. Suppose that S is complete with respect to the induced Riemannian metric. Then S is an entire graph and Ω is simply connected.*

Before laying out the proof let's recall a couple of results from the theory of coverings.

Proposition 2. *Let M and N be two differentiable manifolds and $f : M \to N$ a smooth map*

- *If M is simply connected, N is connected and f is a covering map, then it is a homeomorphism.*
- *We say that the map f satisfies the path lifting property if for every $p \in N$ and for every path $\alpha : I \to M$ such that $\alpha(0) = f(p)$ there exists a path $\tilde{\alpha} : I \to N$ such that $\tilde{\alpha}(0) = p$ and $f \circ \tilde{\alpha} = \alpha$. We can assume without loss of generality that the paths involved are differentiable.*
- *If f is a local homeomorphism and satisfies the path lifting property, then it's a covering map.*

Proof. The idea of the proof is to show that $\Pi \circ S$ is a global diffeomrphism. Clearly this implies that Π restricted to S is a diffeomeorphism on $\mathbb{R}^2$ and so that S is an entire graph.

Let's observe that the target of $\Pi \circ S$ is just $\mathbb{R}^2$, which is simply connected, and the domain is Ω, which is connected since it's a domain. We already proved, by showing that S is locally a graph, that the map $\Pi \circ S$ is a local homeomorphism. This means we only need to show that it satisfies the path lifting property. After that, by applying the proposition above we conclude that it's a diffeomorphism.

Let's fix a point $z \in \Omega$ and a path $\gamma : [0,1] \to \mathbb{R}^2$ of bounded length, $\ell(\alpha) < \infty$, such that $\gamma(0) = \Pi \circ S(z)$. Define then

$$\mathcal{I} := \{t \in [0,1] \mid \exists \gamma^* : [0,t] \to \Omega, \gamma = (\Pi \circ S) \circ \gamma^*, \gamma^*(0) = z\} \quad \text{and} \quad \bar{t} := \sup \mathcal{I}.$$

We will be able to lift γ if we prove that $\bar{t} = 1$.
It's clear to see that $0 \in \mathcal{I}$ and that it's open, since from $\Pi \circ S$ being local homeomorphism we get that $t \in \mathcal{I} \cap (0,1)$ implies $(t - \varepsilon, t + \varepsilon) \subset \mathcal{I}$ for a small ε.
Let's suppose by contradiction that $\bar{t} < 1$.
First let's prove that $\lim_{t \to \bar{t}} \gamma^*(t)$ exists.

- Let's observe that the map Π dilates distances, as

$$\begin{aligned} \langle v, v \rangle_{\mathbb{M}^{2,1}} &= (v^1)^2 + (v^2)^2 - (v^3)^2 \\ &\leq (v^1)^2 + (v^2)^2 \\ &= \langle \Pi v, \Pi v \rangle_{\mathbb{R}^2} \end{aligned}$$

- Then it's clear that $\ell(\gamma^*) \leq \ell(\gamma)$.
- Now, if we fix any sequence $t_k \to \bar{t}$ and observe that as γ is differentiable, it's also Lipschitz, $d(\gamma(t_k), \gamma(t_m)) \leq C|t_k - t_m|$. By observing that

$$\begin{aligned} d(\gamma^*(t_k), \gamma^*(t_m)) &\leq d(\gamma(t_k), \gamma(t_m)) \\ &\leq C|t_k - t_m| \end{aligned}$$

  we deduce that also $\gamma^*(t_k)$ is a Cauchy sequence, hence converging since S is complete and Ω is to be intended with the induced metric.
  By the arbitrariness of the sequence $(t_k)_k$ we deduce the limit $\lim_{t\to\bar{t}} \gamma^*(t)$ exists.

Since the limit exists, we deduce $\bar{t} \in \mathcal{I}$. But now, as the set $\mathcal{I}$ is open, having $\bar{t} \in I$ and $\bar{t} < 1$ implies that $t + \varepsilon \in \mathcal{I}$, which is a contradiction.
We deduce then that $\bar{t} = 1$ and that $\Pi \circ S$ satisfies the lifting property above.

□

It's not difficult to convince ourselves, as the local representation as a graph is always possible, that the conditions of being able to explicit the time variable as dependent and that $|\vec{\nabla} f|^2 < 1$ are a characterization of space-like immersed surfaces.sistemare

The Gauss map in $\mathbb{M}^{2,1}$
The surface S is an immersed submanifold of codimension 1. As we have a non-degenerate form on the ambient space, g, once fixed a $p \in S$, we can decompose the tangent spaces as follows:

$$\mathbb{M}^{2,1} = T_p\mathbb{M}^{2,1} = (T_pS)^\perp \oplus T_pS$$

Having codimension 1, the dimension of $(T_pS)^\perp$ is exaclty one, and in fact it's generated only by the normal vector to S, $\vec{n}$.
We can easily show that n must be a time-like vector, due to the fact that T_pS is made of space-like vectors.
Let's fix $e_1, e_2 \in T_pM$ two independent. As n belongs to $(T_pM)^\perp$ we have $\langle e_1, n\rangle = \langle e_2, n\rangle = 0$. This implies that when we look at the representative matrix of g_p, we have

$$[g_p] = \begin{pmatrix} \langle n, n\rangle & 0 & 0 \\ 0 & \langle e_1, e_1\rangle & \langle e_1, e_2\rangle \\ 0 & \langle e_2, e_1\rangle & \langle e_2, e_2\rangle. \end{pmatrix} =: \begin{pmatrix} \langle n, n\rangle & 0 \\ 0 & [(g|_S)_p] \end{pmatrix}.$$

Where the block $[(g|_S)_p]$ represents the form g on the subspace T_pM. We conclude that $\langle n, n\rangle$ is negative just by observing that $\det[g_p] < 0$ as g is a pseudometric, while $\det[(g|_S)_p] > 0$ since g becomes a metric when restricted to S.

We can now repeat the classical construction of the Gauss map in this pseusoriemannian context.
The main idea is to take a unit vector in $(T_pM)^\perp$, as we have just seen we can't impose the squared norm to be 1, which means we need to normalize it to -1.

Given $w \in (T_pM)^\perp$ we scale by a factor k such that $\langle kw, kw \rangle = -1$. This equation has two different solutions, that send w to one of the two branches of $\hat{\mathbb{H}}^2 = \{v \in \mathbb{M}^{2,1} | \langle v, v \rangle = -1\}$, so we need to make a choice. We consistently choose it to be in $\mathbb{H}^2 = \{v \in \mathbb{M}^{2,1} | \langle v, v \rangle = -1, v^3 > 0\}$.

This way we have constructed a map

$$u : S \to \mathbb{H}^2,$$

the Gauss map of the surface S. Of course the only difference from the classical theory of surfaces is the codomain of the map u, where it is the sphere $\mathbb{S}^2$, and like in the classical case, the choice of the normal vector specifies an orientation for the surface.

$\mathbb{H}^2$ is a model for the Hyperbolic space.

The curvatures of a space-like map
tbd

C Harmonic maps

D Classification of CMC

D.1 The idea of the proof

D.2 The analytical work